

① Grey Level Transformation:

- a) Grey level transformation is used to improve the appearance of an image by changing the intensity values of an pixels. It helps in enhancing contrast, highlighting details and making objects easier to identify.
- b) Negative Transformation

Formula:

$$S = 255 - r$$

original pixel (r)	Calculation	Transform pixel (S)
50	$255 - 50$	205
100	$255 - 100$	155
150	$255 - 150$	105
200	$255 - 200$	55

Interpretation:

The negative transformation inverts the image. Dark pixels becomes bright and bright pixels become dark.

② Histogram Equalization

Histogram Equalization improves image contrast by redistributing intensity values, making dark and bright regions more distinguishable.

Given Histogram

Intensity	Frequency
0	4
1	6
2	10
3	6

Total pixels

$$N = 4 + 6 + 10 + 6 = 26$$

b) Cumulative Distribution Function (CDF)

Intensity	Frequency	CDF
0	4	4
1	6	10
2	10	20
3	6	26

c) Equalized Intensity values

Formula:

$$S = \frac{CDF}{N} \times (L-1)$$

$$L = 4$$

$$L-1 = 3$$

Intensity	CDF	calculation	Equalized value
0	4	$(4/26) \times 3 = 0.46$	0
1	10	$10/26 \times 3 = 1.15$	1
2	20	$20/26 \times 3 = 2.31$	2
3	26	$26/26 \times 3 = 3$	3

Interpretation:

The histogram becomes more evenly distributed

③ Gaussian Smoothing

a) Gaussian filtering is preferred because it reduces image noise while preserving edges better than average filters.

b) Apply Gaussian Kernel

Kernel

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Image

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

$$= \frac{1}{16} [(10) + (40) + (10) + (40) + 160 + 40 + 10 + 40 + 10]$$

$$= 360/16$$

$$= 22.5 = 23$$

Interpretation:

The Centre pixel change from 40 to 23, reducing noise and smoothing image.

④ edge detection:

a) edge detection identifies sudden detection in intensity to locate object boundaries, shapes and important image features.

b) Apply Horizontal Sobel mask $\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$

$$\text{Image} \begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

$$= (-1 \times 10) + (-2 \times 10) + (-1 \times 10) = -40$$

$$= 0$$

$$= (1 \times 30) + (2 \times 30) + (1 \times 30) = 120$$

$$\Rightarrow 120 - 40 = 80 \Rightarrow G = 80$$

Interpretation

The large positive gradient indicates a strong horizontal edge in the image

⑤ Contrast Stretching

a) Need for Contrast Stretching

Contrast Stretching expands making the intensity range of an image, making dark regions darker and bright regions brighter.

b) Linear Contrast Stretching

formula:

$$s = \frac{(r - r_{\min})}{(r_{\max} - r_{\min})} \times 255$$

Given $r_{\min} = 50$
 $r_{\max} = 150$

At $r = 50$ $s = \frac{50 - 50}{100} \times 255 = 0$

At $r = 100$ $s = \frac{100 - 50}{100} \times 255 = 0.5 \times 255$
 $= 127.5$
 $= 128$

At $r = 150$ $s = \frac{150 - 50}{100} \times 255 = 255$

original Intensity	New Intensity
50	0
100	128
150	255

Interpretation:

The Intensity range is stretched from 50-150 to 0-255, resulting the higher contrast and clear image.